

KUSHAGRA AGARWAL

Full Stack Engineer

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Professional Experience

Team Conatus (Student Technical Society)

Aug 2024 – Jul 2025

Full Stack Developer — MERN Stack

Ghaziabad, India

- Developed and deployed Full Stack event registration platforms for 4 college-wide technical events using the MERN stack (MongoDB, Express.js, React, Node.js), owning end-to-end feature delivery from UI to database
- Built responsive React frontends with dynamic form handling, real-time validation, and seamless REST API integration — ensuring smooth user experience across devices for event participants
- Designed and implemented RESTful backend APIs with Node.js and Express.js, including JWT-based authentication and MongoDB schema design for managing event registrations and user data
- Mentored and guided a team of 5–7 junior developers in MERN stack best practices, code reviews, and debugging workflows, accelerating feature delivery and onboarding new contributors

Key Projects

GaganDrishti — AWS Cloud Security Automation Platform | Node.js, Express, MongoDB, AWS [GitHub](#) 2026

- Architected an AWS cloud security scanner implementing 9 detection rules across S3, EC2, IAM, and RDS, classifying findings into a 4-tier severity model (CRITICAL to LOW) covering exposed storage, open ports, over-permissive IAM policies, and encryption gaps.
- Implemented an asynchronous scan orchestration engine using Node.js, BullMQ, and Redis supporting 5 concurrent scan jobs with automatic retry (3 attempts, exponential backoff) for AWS rate-limit resilience, alongside AES-256 encrypted vaulting of IAM credentials in MongoDB.
- Engineered a React/Vite dashboard and RESTful APIs featuring real-time WebSocket (Socket.io) job tracking across a 4-phase scan lifecycle, supporting both full-environment and targeted scan modes with actionable, rule-specific remediation guidance per finding.

AutoCrawler — Web Automation System | Node.js, React, MongoDB, Socket.IO, Cheerio, n8n [GitHub](#) 2026

- Built a Full Stack web scraping and automation platform with a hybrid crawler that auto-detects JS frameworks (React, Vue, Angular, Next.js) to choose between lightweight HTTP requests and headless-browser (Puppeteer) rendering
- Developed a scalable backend with configurable rate limiting (500–5000ms request delay), recursive crawling up to 5 levels deep, and retry logic with exponential backoff (up to 3 attempts) for webhook delivery reliability
- Implemented real-time crawl monitoring via WebSockets, indexed MongoDB storage, paginated REST APIs (up to 100 results/page), n8n webhook integrations, and automated recurring crawl workflows

InsightIQ | React, TypeScript, Node.js, PostgreSQL, Gemini AI, Recharts [GitHub](#) 2025

- Developed a full-stack AI-powered analytics platform converting natural language to SQL using the Gemini 2.5 Flash API, with automated query correction, schema-aware validation, and a default LIMIT 100 safeguard against large result sets
- Built a scalable Node.js backend with per-session conversational context tracking (retaining up to 20 messages, using the last 5 queries for follow-up context), a rule-based chart-recommendation engine selecting from 6 chart types based on row count and column data types, and multi-tenant database session management
- Engineered a responsive React + TypeScript frontend with dynamic visualizations enabling real-time, AI-driven insights

Technical Expertise

Programming Languages: JavaScript, TypeScript, Python, C, C++, SQL, HTML/CSS, Bash

Backend Development: Node.js, Express.js, RESTful API Design, JWT Authentication, WebSocket (Socket.IO), Middleware Architecture, Async Job Queues (BullMQ, Redis)

Frontend Development: React.js, Tailwind CSS, Responsive Design, Real-time Updates (Socket.IO), REST API Integration

Databases: MongoDB, PostgreSQL, MySQL, Mongoose ODM, Database Indexing, Query Optimization, Schema Design

Cloud & Security: AWS Cloud Security (S3, EC2, IAM, RDS), IAM Configuration Analysis, Cloud Resource Scanning, Credential Encryption, Vulnerability Assessment, Input Validation, Rate Limiting

DevOps & Automation: n8n Workflow Automation, Webhook Integration, Scheduled Job Processing, CI/CD Fundamentals, Docker (Basics), Linux (Debian)

Tools & Methodologies: Git, GitHub, Postman, MVC Architecture, Web Architecture Design, REST Standards, WebSocket Architecture, Agile Workflow, Data Structures & Algorithms

Core Competencies: Problem Solving, Team Collaboration & Mentorship, Organizational Skills, Verbal & Technical Communication

Education

Dr. A.P.J. Abdul Kalam Technical University

Aug 2023 – May 2027 (Expected)

Bachelor of Technology, Computer Science and Engineering – CGPA: 8.8/10.0

Ghaziabad, India